Towards a Bernstein-von Mises theorem for Bayesian inference with data missing at random

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Missing data

- Missing covariates are common in practice.
- Bayesian procedures naturally integrate over missing values instead of plugging in a single completed dataset.
- But when do posterior credible sets still have frequentist meaning?

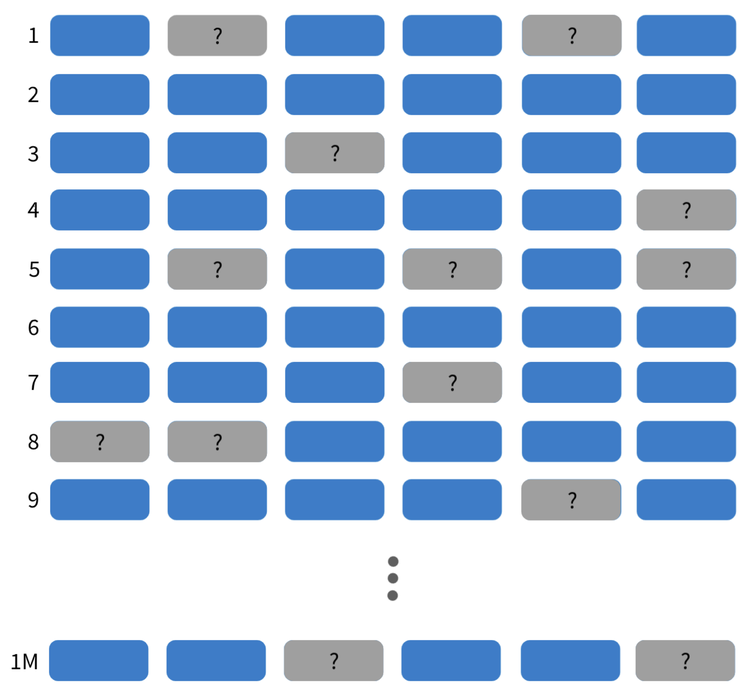


Figure 1: Missing data, source: `https://towardsdatascience.com/how-to-handle-missing-data-b557c9e82fa0/`

Missing at random

If the complete data is X, and the observed data is X_{obs}, then the missing data mechanism is missing at random (MAR) if

$$\mathbb{P}\left(\text{Missingness pattern} = m \mid |X = x\right) = f(m, x_{\text{obs}}),$$

for some function f that does not depend on the missing values x_{mis}.
Interpretation: the probability of missingness depends only on the observed data, not on the missing data.

Example problem

$$Y = \beta^T X + \epsilon, \quad \epsilon \sim N(0, \sigma^2), \quad X \sim f_0$$

Goal, estimate β, while (X, Y) is missing at random.

Obstacle

f_0 is unknown, and thus we have to model it. But if we model f_0 incorrectly, then the estimates of β can be suboptimal.

Solution

We use a nonparametric prior on f_0, which allows us to estimate β without making strong assumptions on the distribution of X.

Problem:

When is our Bayesian procedure valid? When are posterior credible sets valid frequentist confidence sets?

Main goal

We want to prove the following Bernstein-von Mises theorem for β:

Theorem 1. *Under some regularity conditions,*

$$\mathbb{P}\left(\sqrt{n}(\beta - \hat{\beta}) \in \cdot \mid \mathit{Data}\right) \approx N(0, I_{\mathit{eff}}^{-1}),$$

where $\hat{\beta}$ is the efficient estimator of β_0, and I_{eff} is the efficient information matrix.

Implications

- The posterior distribution of β is asymptotically normal, centred at the efficient estimator $\hat{\beta}$.
- The posterior credible sets for β are asymptotically equivalent to frequentist confidence intervals.
- This result holds, even when the distribution of X is unknown and modelled nonparametrically.

Intuition

We infer β by imputing the missing covariates using the posterior for f, and averaging over all plausible imputations. When the posterior for f concentrates around f_0, the imputations are close to the true distribution of X, so the posterior for β remains valid and accounts for imputation uncertainty.

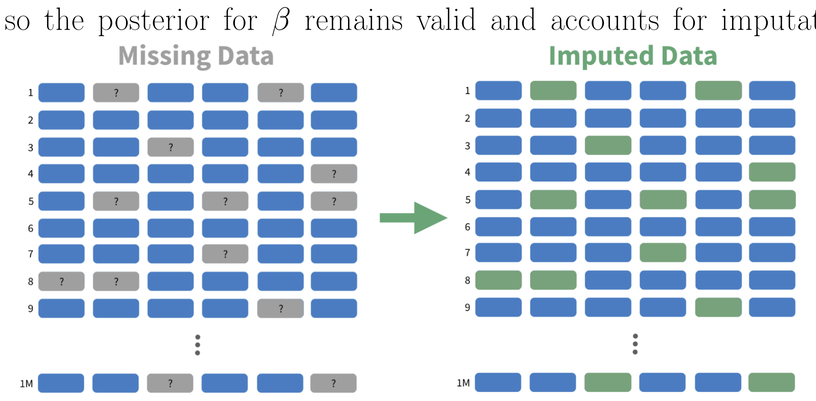


Figure 2: Imputation of missing values, source: `https://towardsdatascience.com/how-to-handle-missing-data-b557c9e82fa0/`

Examples of priors we expect to handle:

- log-spline priors,
- log-wavelet priors,
- histogram priors,
- log-Gaussian process priors.

Current status

- **LAN expansion:** essentially known / expected to follow from the current approach.
- **Change-of-measure condition:** main technical step currently being proved.
- **Final BvM theorem:** Standard approach, but requires the previous two steps to be completed and final details filled in.

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References

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2. Franssen, S., Nguyen, J. & van der Vaart, A. *The Bernstein-von Mises Theorem for Semiparametric Mixtures* Nov. 2024. arXiv: `2412.00219 [math]`.
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